1.4 Getting Started

You can control the robotic arm through 6-channel knob expansion board. Due to 5 degrees of freedom, only knobs S1~S2 can control the MiniArm. Knob S6 is ineffective.

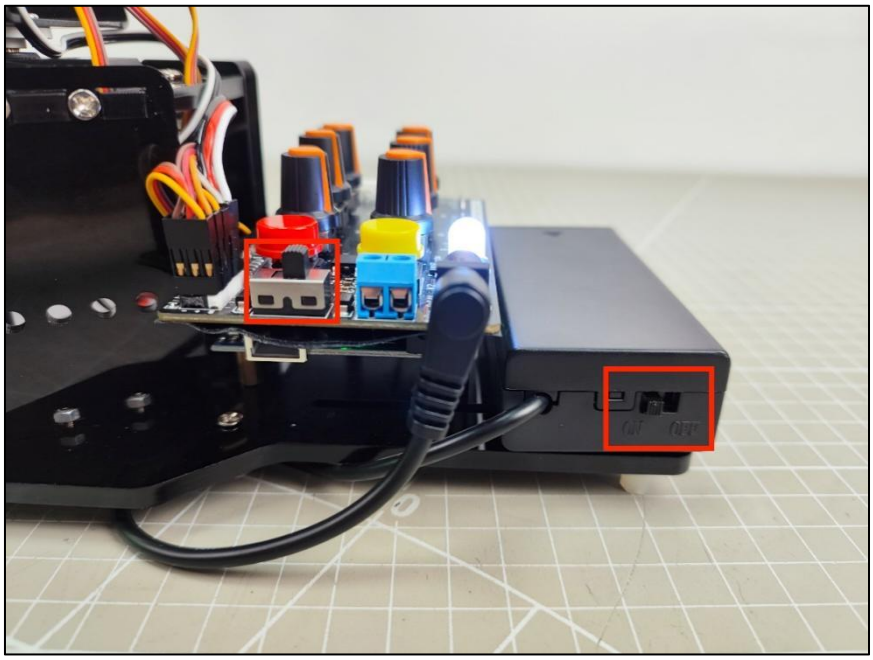
1. Preparation and Notice

1）Ensure that the robotic arm is installed correctly and the servo wire is connected correctly. You can learn how to assembly and wiring in “1. Getting Ready/1.2 Wiring and Charging Instructions”.

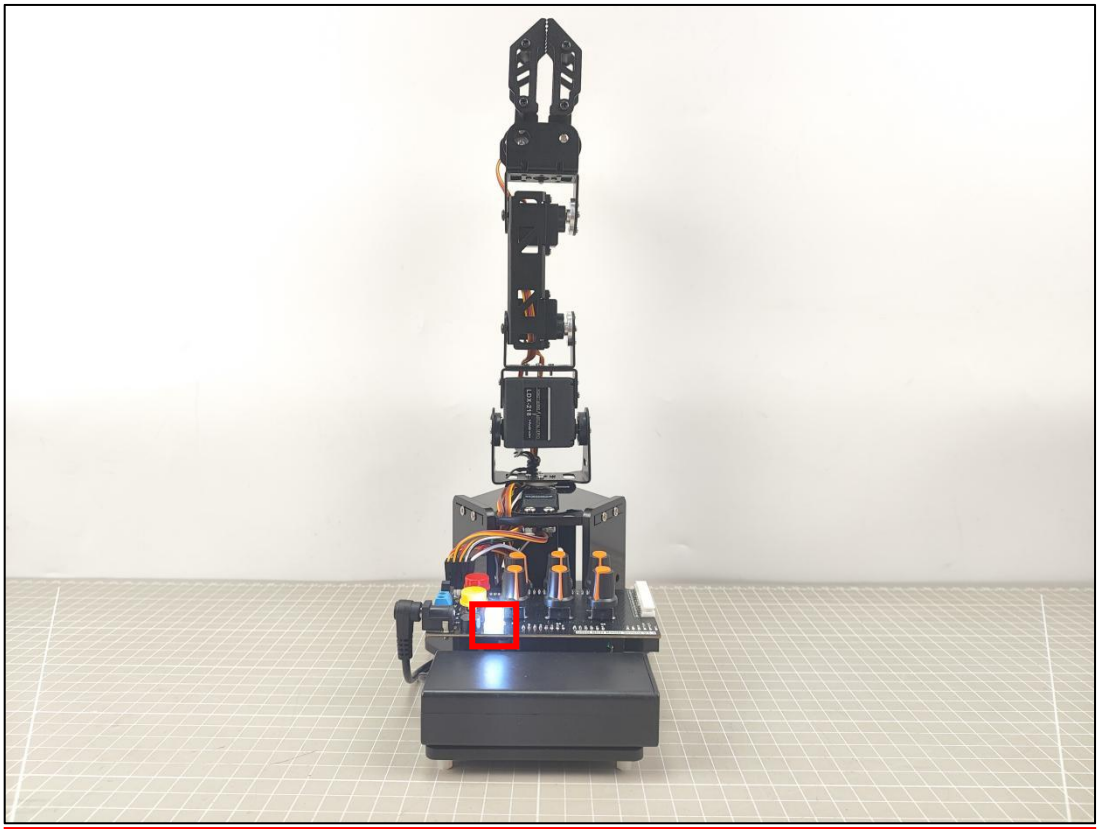
2）If the servo cannot rotate normally when turning the knob, please do not continue turning the knob to avoid servo blocking.

2. Power On

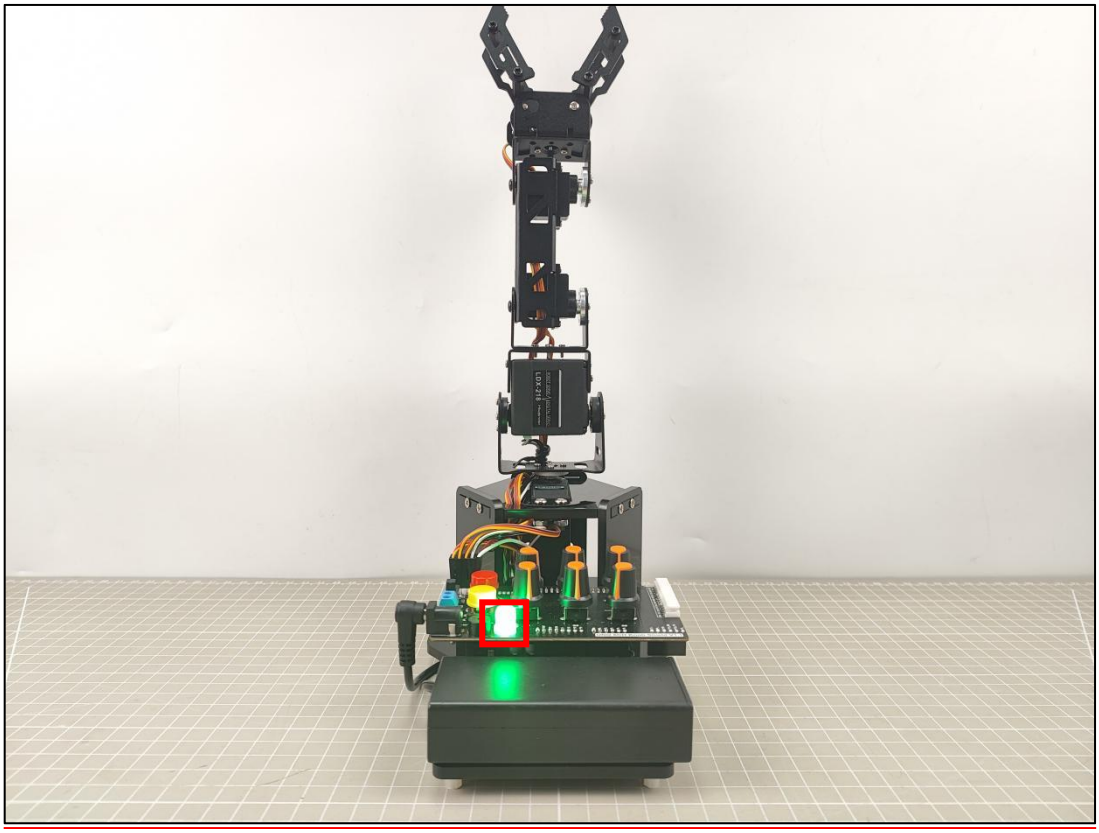
1）Turn the switch of the battery box and expansion board to “ON”.



2)The RGB on the expansion board will light up white, and the robotic arm will return to the initial position.



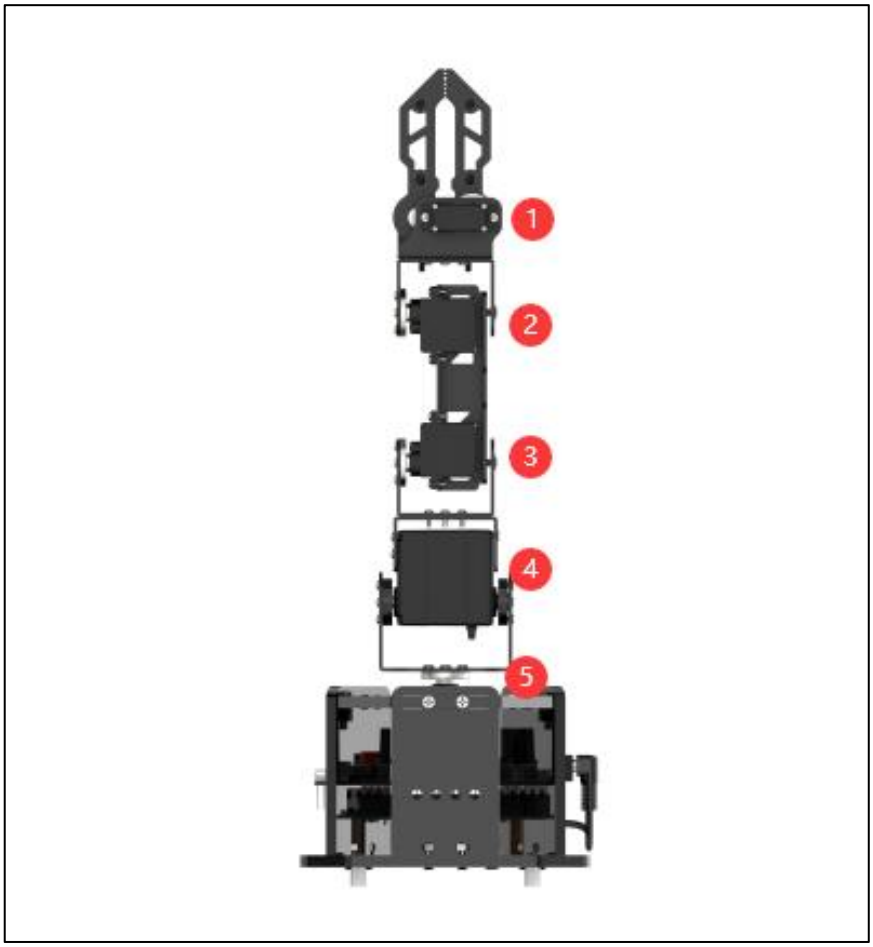
3)Press K1 and K2 buttons on the expansion board at the same time, then release to exit the neutral mode until RGB light up green. You can not control by knob or app if you do not exit the neutral mode.



3. Knob Control Steps

After exiting the neutral mode for two seconds, the servos on the robotic arm will rotate to the position the current knob in. You can rotate knobs S1~S5 on the expansion board to control servo No.1-5 respectively.

1)The correspondence between knobs and servos is as follows:



Knob	Servo	Control Instruction (robotic arm)
S1	Servo No.1 (claw)	Knob controls the opening and closing of the claw
S2	Servo No.2 (wrist joints)	Knob controls the rotation of each joint servo
S3	Servo No.3 (forearm)	
S4	Servo No.4 (arm)	
S5	Servo No.5 (pan-tilt)	Knob controls pan-tilt rotation

2)In addition to the expansion board described above, you can control the

robotic arm by other modes.

For example, the RGB on the expansion board will light up red to enter editing action group mode when long pressing the red button on the expansion board. After the edition is completed, the RGB will switch to yellow to indicate the action group is executed.

When connecting to the robotic arm to app, the RGB light on the expansion board will switch to blue to proceed app control mode. The correspondence between RGB lights and modes can be referred as follows:

RGB Light Color	Mode	Function
white	neutral mode	Servo ID 1~5 are in the initial position
green	normal mode	knob control
red	action group editing mode	editing action
yellow	action group running mode	Run the offline action group edited in red light
blue	App mode	Control robotic arm through App

You can learn how to specifically edit a complete action group in “2. Action Editing/2.1 Offline Knob Programming”.

You can learn how to use phone to connect and control robotic arm in “3. Remote Control Course/3.1 APP Control”.